

CSCE 632 ACCESSIBLE COMPUTING · MODULE 12 · LECTURE NOTES

Why It Fails

The organisational causes, which are the real ones.

MODULE OUTCOMES

- Explain why accessibility fails organisationally.
- Make the cost argument correctly.
- Explain the retrofit cost and its cause.
- Explain what actually produces change.
- Specify a process that holds accessibility.

1 The failure pattern

ACCESSIBILITY FAILS AS A LATE-STAGE CHECK AGAINST WORK THAT WAS ALREADY FINISHED

The design is approved, the implementation is built, the release date is set, and then an accessibility audit arrives — at which point every single finding is a change request against finished work, competing directly with features for the remaining time.

So the findings are triaged, the structurally serious ones are deferred because they are expensive, and the cheap ones are fixed — producing an interface with correct alt text and an unusable focus order, which is the characteristic outcome and is recognisable in a great deal of shipped software.

And nobody involved made a bad decision: each triage call was locally reasonable given the schedule and the alternatives — which is precisely why exhortation does not fix this and process does, and is the same structure as CSCE 713 Module 11's incident analysis.

Which means the cause is the position in the schedule rather than the priority — the same findings arriving as design feedback would simply have been built that way, at no additional cost whatsoever (§3).

2 The cost argument

- The quality argument works — 'this is a defect that excludes users' is a bug report, and bug reports have an existing process with an owner and a queue (Module 01 §1's social-model consequence).
- The architecture argument works — 'this is cheap now and expensive later' is a technical-debt claim, which engineering leads already understand and already act on for other reasons (§3).

- **The legal argument works, and works poorly — it produces conformance rather than access, which is Module 07 §3's gap turned into a deliberate strategy:** meet the criteria, claim the level, change nothing a user would notice.
- **The market-size argument backfires — it invites arguing about the number** (Module 02 §4's warning), **and the number is genuinely contestable**, so the discussion moves from a requirement to an estimate.
- **And the moral argument is true and does not allocate — everybody already agrees, and agreement is not what is missing. So an argument that produces agreement produces nothing; what is missing is a position in the process**, which is §4's subject.

3 The retrofit cost

ACCESSIBILITY IS NEARLY FREE IN DESIGN AND EXPENSIVE IN REPAIR, AND THE RATIO IS ARCHITECTURAL

Choosing a native element costs nothing; replacing an established custom component library with accessible equivalents costs quarters of engineering time — and the second is the same decision made later (Module 08 §2), which is the clearest instance of the pattern.

Abstracting the input layer behind named actions costs a day in week one and a rewrite in year two (Module 11 §2) — which is the same shape, in a different domain, and with the same cause.

And a design that assumed fixed pixel positions cannot be made to reflow without redoing the layout entirely — so the 400% reflow criterion is effectively decided by a visual design decision nobody labelled as an accessibility decision (Module 03 §4).

Which is the argument to make and is also the one that happens to be true: the cost is not the accessibility, it is the lateness — and that reframes the request from 'additional work' to 'avoided rework', which is a category engineering organisations have a budget for.

4 What changes things

PROCUREMENT

a buyer that requires conformance moves suppliers, which is why public-sector procurement rules have had more effect than any guideline ever published

LITIGATION AND REGULATION

effective, slow, and it produces the conformance-not-access outcome (Module 07)

A DEFECT PROCESS THAT ACCEPTS THESE AS DEFECTS

the single most effective internal change, because it routes the work through machinery that already works

CI GATES

cheap, narrow (Module 09 section 1), and they prevent regression, which is most of the long-run value

DISABLED STAFF, AND USERS WITH A ROUTE IN

which is what changes design rather than only fixing defects -- and is why the games field moved (Module 11 section 4)

Routing accessibility through the existing defect process is the single most effective internal change — because that machinery already works: it has owners, severities, queues, and a culture of not shipping with known serious bugs. An accessibility finding filed as an accessibility finding goes into a separate list

that nobody owns; the same finding filed as a defect competes on equal terms and usually wins. **The defect-process point is the module's practical recommendation**, and it is available to an individual engineer without any authority.

AND THE HONEST SUMMARY OF THIS MODULE

The technical problems in this course are solved — every requirement in Modules 03 through 11 has a known, documented, implemented solution — and the subject's actual failure is organisational rather than technical, which is unusual and worth sitting with.

Which is exactly why this module exists in an engineering course: knowing the fix is not sufficient if the fix arrives after the decision that made it expensive (§3), and most of this course's content arrives at that point by default.

And the leverage is in the schedule and the process — design review, the defect process, the CI gate, and a route for users to report problems — rather than in anybody's conviction, which is already sufficient (§2).

So the thing to carry out of here is a sequence: argue from quality and architecture, get it into the defect process, gate the regressions in CI, and ask the question in design review — which is four moves and none of them require authority, which is what makes them worth knowing as an individual contributor.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Lazar, Goldstein & Taylor — Ensuring Digital Accessibility through Process and Policy	The whole module — the organisational and policy causes, with case studies. Library copy.
The W3C's planning and managing accessibility resources (free)	§4 — a process model, including the design-review placement this module argues for.
EN 301 549, the European accessibility procurement standard (free)	§4's procurement lever , in the form a supplier actually encounters it.
The WebAIM Million trend data (free)	§1's outcome, measured over years — and the slow improvement is the evidence that this module's pessimism is warranted.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Describe the failure pattern** from a project you have seen.
- 2 **Find an interface with good alt text and a bad focus order**, and infer its history.
- 3 **Write the same finding five ways**, one per argument in §2.
- 4 **File one accessibility finding as an ordinary defect** and note what happens.
- 5 **Estimate the retrofit cost** of making one existing component library accessible.
- 6 **Estimate what it would have cost in week one**.
- 7 **Find a pixel-positioned layout** and estimate the reflow cost.
- 8 **Read one procurement accessibility requirement** in full.
- 9 **Add an automated accessibility gate** to one CI pipeline.
- 10 **Write the design-review question** you would add, in one sentence.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 State the failure pattern and its characteristic outcome.
- 2 Why does blaming the participants not help?
- 3 What is the actual cause?
- 4 Rank the five arguments, with reasons.
- 5 Why does the market-size argument backfire?
- 6 Why is the moral argument insufficient despite being true?
- 7 Give three instances of the retrofit ratio.
- 8 State the reframing that the timing argument allows.
- 9 Name five things that change organisations.
- 10 Which internal change is most effective, and why?